

## Ch. 5 Notes

### 5.1 Cross Sections

cross sect. Area

$$\sigma = \frac{\# \text{ particles}}{\text{time} \times \# \text{ density} \times \text{vol beam}}$$

$$= \frac{1}{T} \frac{1}{I} N, \quad I = \# \text{ density} \times \text{vol (beam)}$$

- Natural to measure diff. cross sec.  $\frac{d\sigma}{d\Omega}$   
change in angle

- Gives info (classically)  
about shape of object

Q11

Generalize to cross section

- QM has prob. of scattering

$$P = \frac{N}{N_{\text{inc}}} \text{ of } N_{\text{inc}} = \# \text{ incident particles}$$

cross section then

$$d\sigma = \frac{1}{T} \frac{1}{I} dP \text{ of } \Psi \text{ flux normalized}$$

diff # of scatterings is

$$dN = L \times d\sigma \text{ w/ } L = \text{length}$$

Cross section for S matrix elements  
(1i) two particle state

i.e.  $2 \rightarrow n$  process

$$p_1 + p_2 \rightarrow \{p_j\}$$

$$\text{Intr: } \mathcal{I} = \frac{|\vec{v}|}{v} \left( \text{but } 2 \text{ particles} \right. \\ \left. \frac{|\vec{v}_1 - \vec{v}_2|}{v} \right) \\ \Rightarrow d\sigma = \frac{v}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} d\mathcal{P}$$

"BF just a bunch of  $\delta$  functions"

$$d\mathcal{P} = \frac{|\langle f | S | i \rangle|^2}{\langle f | f \rangle \langle i | i \rangle} d\pi \quad \text{w/ } d\pi \text{ 'region of final state} \\ \text{momenta at which we are looking'}$$

$$d\pi = \prod_j \frac{N}{(2\pi)^3} d^3 p_j$$

$$\text{w/ } \int d\pi = 1 \quad \text{because } \left( \frac{d\mathcal{P}}{2\pi} \right) = \frac{1}{L}$$

$$a_\mu^\dagger(0) = \frac{1}{\sqrt{2\omega_\mu}} |u\rangle$$

$$[a_p, a_q^\dagger] = (2\pi)^3 \delta^3(p-q)$$

$$\hookrightarrow \langle p | p \rangle = -(2\pi)^3 (2\omega_p) \delta^3(0)$$

$\delta^3(\mathbf{p})$  infinite but

$$(2\pi)^3 \delta^3(\mathbf{p}) = \int d^3\mathbf{u} e^{i\mathbf{p} \cdot \mathbf{u}}$$

$$\text{so } \delta^3(\mathbf{p}) = \frac{1}{(2\pi)^3} \int d^3\mathbf{u} = \frac{V}{(2\pi)^3}$$

$$\dagger \delta^4(\mathbf{p}) = \frac{TV}{(2\pi)^4}$$

$$\Rightarrow \langle p | p \rangle = 2\omega_p V = 2E_p V$$

$$\text{since } |i\rangle = |p_i\rangle |f\rangle, \quad f = \pi |p_j\rangle$$

$$\langle i | i \rangle = (2E_1 V)(2E_2 V), \quad \langle f | f \rangle = \pi (2E_2 V)$$

- calc S matrix perturbation by (lowest order is just  $\mathbb{1}$  since no interactions)

$$S = \mathbb{1} + i\mathcal{T} \quad (i, j \text{ just convention since } S \sim e^{i\hat{T}})$$

$\mathcal{T}$  is transition matrix

- equal & converse unless  $H^0$  &  $|i\rangle$  have same  $\mathbf{p}$ .

$$\mathcal{T} = (2\pi)^4 \delta^4(\mathbf{E}_f) \mathcal{M} \quad \text{w/ } \delta^4(\mathbf{E}_f) = \delta^4(p_{f_1}^0 - p_{f_2}^0)$$

Matrix elements

$$\langle f | S - \mathbb{1} | i \rangle = i(2\pi)^4 \delta^4(\mathbf{E}_f) \langle f | \mathcal{M} | i \rangle$$

$$\rightarrow |\langle P|S|i\rangle|^2 = g^4(\epsilon_P) g^4(\Sigma P) (2\pi)^8 |\langle P|M|i\rangle|^2 \\ = g^4(\Sigma P) T L (2\pi)^4 |M|^2$$

→ phase

$$dP = \frac{g^4(\epsilon_P) T L (2\pi)^4}{(2E_1 L)(2E_2 L)} \frac{1}{\pi_j (2E_1 L)} |M|^2 \frac{T L}{(2\pi)^3} d^3 p_j \\ = \frac{T}{L} \frac{1}{(2E_1)(2E_2)} |M|^2 d\pi_{LIPS}$$

w/  $d\pi_{LIPS}$  (2d invariant phase space)

$$\equiv \prod_{\substack{\text{final} \\ \text{states}}} \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_{p_j}} (2\pi)^4 g^4(\epsilon_P)$$

$$\Rightarrow d\sigma = \frac{1}{(2E_1)(2E_2) |\vec{v}_1 - \vec{v}_2|} |M|^2 d\pi_{LIPS}$$

Decay rates:

$$d\Gamma = \frac{1}{T} dP \quad (\text{prob } P, \text{ turns into } i \text{ j's over time } T)$$

- can't use S-matrix to describe decays

$$\rightarrow d\Gamma = \frac{1}{2E_i} |M|^2 d\pi_{LIPS}$$

spectral cases

$2 \rightarrow 2$

$$p_1 + p_2 \rightarrow p_3 + p_4$$

$$\vec{p}_1 = -\vec{p}_2, \vec{p}_3 = -\vec{p}_4 \Rightarrow E_1 + E_2 = E_3 + E_4 = E_{cm}$$

$$d\Gamma_{LIPS} = (2\pi)^4 \delta^4(\Sigma p) \frac{d^3 p_3}{(2\pi)^3} \frac{1}{2E_3} \frac{d^3 p_4}{(2\pi)^3} \frac{1}{2E_4}$$

$$d\Gamma_{LIPS} = \frac{1}{16\pi^2} d\Omega \int d p_f \frac{\vec{p}_f^2}{E_3} \frac{1}{E_4} \theta(E_3 + E_4 - E_{cm})$$

$$|p| = |\vec{p}_3| = |\vec{p}_4|, E_3 = \sqrt{m_3^2 + p^2}, E_4 = \sqrt{m_4^2 + p^2}$$

change from  $p_f$  to  $p(p) = E_3(p) + E_4(p) - E_{cm}$

$$\frac{dp}{dp_f} = \frac{d}{dp_f} (E_3 + E_4 - E_{cm}) = \frac{p_f}{E_3} + \frac{p_f}{E_4} = \frac{E_3 + E_4}{E_3 E_4} p_f$$

since  $E_3 + E_4 = E_{cm}$

$$d\Gamma_{LIPS} = \frac{1}{16\pi^2} d\Omega \int_{m_3+m_4-E_{cm}}^{\infty} d p \frac{p_f}{E_{cm}}$$

$$?? = \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{cm}} \theta(E_{cm} - m_3 - m_4)$$

$$d\sigma = \frac{1}{(2E_1 2E_2) |\vec{v}_1 - \vec{v}_2|} \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{cm}} |M|^2 \theta(E_{cm} - m_3 - m_4)$$

since  $|\vec{v}_1 - \vec{v}_2| = \left| \frac{\vec{p}_1}{E_1} + \frac{\vec{p}_2}{E_2} \right| = |\vec{p}_1| \frac{E_{cm}}{E_1 E_2}$

$$so \left( \frac{d\sigma}{d\Omega} \right)_{CM} = \frac{1}{64\pi^2 E_{cm}^2} |M|^2 \quad (\text{under on-shell masses approx})$$

## 3.2 Non Relativistic Limit

Born Approx

(electron  $\phi_e$  and  $\mu$  scatter off proton  $\phi_p/\mu_p$ )

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} \left[ \tilde{V}(\vec{k}) \right]^2$$

$$\tilde{V}(\vec{k}) = \int d\vec{r} e^{i\vec{k}\cdot\vec{r}} V(\vec{r})$$

$\mu$  diff. of electron momentum  
(momentum transfer)

$$\text{if } V(r) = \frac{e^2}{4\pi|\vec{r}|} \text{ then } \tilde{V}(\vec{k}) = \frac{e^2}{k}$$

$$\rightarrow \left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} \left(\frac{e^2}{k}\right)^2$$

$$\text{Have } \left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{1}{64\pi^2 k_p^2} |M|^2$$

because  $E_p \propto k_p$

Dimension of  $M$  :

$$\left[\frac{d\sigma}{d\Omega}\right] = -2$$

$$[m_p^{-2}] = -2$$

$\Rightarrow M$  dimensionless

h.c.

proton & photon Lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 - \phi_e^\dagger (\square + m_e^2) \phi_e - \phi_p^\dagger (\square + m_p^2) \phi_p \\ - ie A_\mu (\phi_e^\dagger \partial_\mu \phi_e - \phi_e \partial_\mu \phi_e^\dagger) \\ + ie A_\mu (\phi_p^\dagger \partial_\mu \phi_p - \phi_p \partial_\mu \phi_p^\dagger) + \mathcal{O}(e^2)$$

$$\text{or} \\ -ie A_\mu (\phi_e^\dagger \overleftrightarrow{\partial}_\mu \phi_e)$$

$$+ ie A_\mu (\phi_p^\dagger \overleftrightarrow{\partial}_\mu \phi_p)$$

(scalar QED Lagrangian)

non-rel. limit

$p^\mu = (E, \mathbf{p})$  close to rest of (mass) so

$E \sim m$  ( $\partial_t \phi \sim i m \phi$ )

$\vec{p} \ll m$

$m/|\vec{p}| \gg 1 \gg m$

$$\text{use } \phi_e \rightarrow e^{i k_e x} \phi_e, \quad \phi_p \rightarrow \phi_p e^{i k_p x}$$

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} F_{\mu\nu}^2 + \phi_e^\dagger \vec{\nabla} \phi_e \\ & + 2emc \phi_e^\dagger \phi_e \\ & + \phi_p^\dagger \vec{\nabla} \phi_p - 2emc \phi_p^\dagger \phi_p \\ & + \dots \text{ (higher order in } \frac{\vec{\nabla}^2}{m^2} \text{)} \end{aligned}$$

Note: matter elements have precise  
proton-photon interaction

(i.e.  $-2em_p$ ), precise electron photon interaction  
( $2em_e$ ),

&  $\frac{1}{k^2}$  propagator from photon line between

$$\therefore M \sim (-2em_p) \frac{1}{k^2} (2em_e)$$

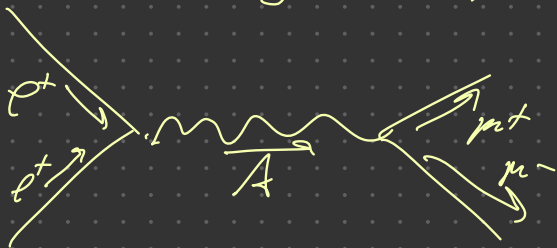
$$\therefore \left( \frac{d\sigma}{d\Omega} \right)_{\text{cm}} = \frac{1}{64\pi^2 m_p^2} |\mathcal{M}|^2 \sim \frac{e^4 m_e^2}{4\pi^2} \frac{1}{k^4}$$

- non rel. limit

$\therefore$  standard to rescale  $\mathcal{L}$  by  $d \rightarrow \frac{\Phi}{\sqrt{2m}}$   
similar to  $\frac{1}{2m_p}$  factor)

$$e^+ e^- \rightarrow \mu^+ \mu^-$$

Diagram: (checking order.)



- $e^+$  &  $e^-$  annihilate into virtual photon
- photon splits into  $\mu^+ \mu^-$  pair (decays)

Amplitudes, let's see

Com Frame:

propagator is

$$\frac{1}{m^2} = \frac{1}{E_{cm}^2}$$

$\nearrow$  in denominator, cancel

$E_{cm}$  cancelled by factor of  $\sqrt{2E_1} = \sqrt{2E_2} = \sqrt{2E_3} = \sqrt{2E_4}$

normalization  $= \sqrt{E_{cm}}$

only remaining part of  $M$  provided by  
 spins into photon polarizations  
 initial

$$M(q_1, q_2 \rightarrow \epsilon_3, \epsilon_4) = \sum_{\lambda} (q_1, q_2 | \epsilon) (\epsilon | \epsilon_3, \epsilon_4)$$

—  $\epsilon$  photon polarizations

- $q_1, q_2$  incoming spins
- $\epsilon_3, \epsilon_4$  outgoing states

(we direction  $\vec{z}$  arbitrary)

$e^-, e^+$  momenta:

$$q_1^\mu = (E, 0, 0, E), p_2^\mu = (E, 0, 0, -E)$$

Now rel:

— the state of spin as up & down

rel/QFT:

- think of electrons being polarized like photon  
 helicity
- can have circularly polarized (left & right) or  
 linearly polarized  
 — linear pol. — helicity like polarized  
 right

possible initial spin states:

$$|s, s_z\rangle = |\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$$

since photon spin 1:

electron & positron have to be polarized same direction  
 $(s = 1/2) \quad (s = 1/2)$

possible output polarizations:

$$\epsilon^1 = (1, 0, 0, 0)$$

$$\epsilon^2 = (0, 0, 1, 0)$$

$|\uparrow\uparrow\rangle$  producing  $\epsilon^1$  (only in dir.)  
 $|\downarrow\downarrow\rangle$  producing  $\epsilon^2$  (only in dir.)

Muons similar, (but spin not going in

$$\hat{z} \quad ? \quad ?$$

$$p_3^\mu = E(1, 0, \sin\theta, \cos\theta)$$

$$p_4^\mu = E(1, 0, -\sin\theta, -\cos\theta)$$

- (0) about  $\hat{p}^-$  axis

- (a) about  $\hat{z}$  axis which = 0 due to cylind. symmetry

polarizations:

$$\bar{\epsilon}^1 = (0, 1, 0, 0) \quad \bar{\epsilon}^2 = (0, 0, \cos\theta, -\sin\theta)$$

Matrix elements =

$$\mu_1 = M(|g\rangle \rightarrow |f\rangle) = \xi^1 \bar{\xi}_1 + \xi^1 \bar{\xi}_2 = -1$$

$$\mu_2 = M(|\uparrow\uparrow\rangle \rightarrow |f\rangle) = \xi^2 \bar{\xi}_1 + \xi^2 \bar{\xi}_2 = -\cos\theta$$

→ should average over initial spins

$$\rightarrow |\mu|^2 = |\mu_1|^2 + |\mu_2|^2 = (1 + \cos^2\theta)$$

$$\rightarrow \frac{d\sigma}{d\Omega} = \frac{e^4}{64\pi^2 E_{cm}^2} (1 + \cos^2\theta)$$

(predictable using QFT machinery)